GATE 2014 Set 2 - Electronics and Communication Engineering

Answers are from the official answer key supplied with the paper. Worked solutions are given where available.

GATE 2014 Set 2 | Q.01

Q.1 Choose the most appropriate word from the options given below to complete the following sentence.

Communication and interpersonal skills are _____ important in their own ways.

(A) each (B) both (C) all (D) either

Answer: B

GATE 2014 Set 2 | Q.02

Q.2 Which of the options given below best completes the following sentence?

She will feel much better if she _____.

(A) will get some rest
(B) gets some rest
(C) will be getting some rest
(D) is getting some rest

Answer: B

GATE 2014 Set 2 | Q.03

Q.3 Choose the most appropriate pair of words from the options given below to complete the following sentence.

She could not _____ the thought of _____ the election to her bitter rival.

(A) bear, loosing (B) bare, loosing (C) bear, losing (D) bare, losing

Answer: C

GATE 2014 Set 2 | Q.04

Q.4 A regular die has six sides with numbers 1 to 6 marked on its sides. If a very large number of throws show the following frequencies of occurrence: 1 → 0.167; 2 → 0.167; 3 → 0.152; 4 → 0.166; 5 → 0.168; 6 → 0.180. We call this die

(A) irregular (B) biased (C) Gaussian (D) insufficient

Answer: B

GATE 2014 Set 2 | Q.05

Q.5 Fill in the missing number in the series.

2 3 6 15 ___ 157.5 630

Answer: 45 to 45

GATE 2014 Set 2 | Q.06

Q.6 Find the odd one in the following group

Q,W,Z,B B,H,K,M W,C,G,J M,S,V,X

(A) Q,W,Z,B (B) B,H,K,M (C) W,C,G,J (D) M,S,V,X

Answer: C

GATE 2014 Set 2 | Q.07

Q.7 Lights of four colors (red, blue, green, yellow) are hung on a ladder. On every step of the ladder there are two lights. If one of the lights is red, the other light on that step will always be blue. If one of the lights on a step is green, the other light on that step will always be yellow. Which of the following statements is not necessarily correct?

(A) The number of red lights is equal to the number of blue lights
(B) The number of green lights is equal to the number of yellow lights
(C) The sum of the red and green lights is equal to the sum of the yellow and blue lights
(D) The sum of the red and blue lights is equal to the sum of the green and yellow lights

Answer: D

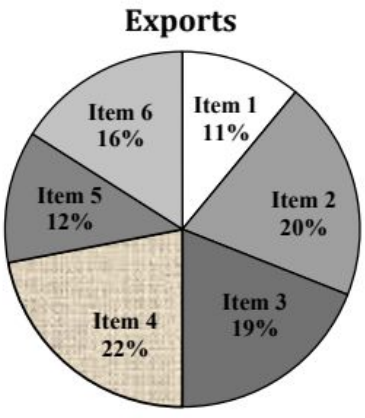
GATE 2014 Set 2 | Q.08

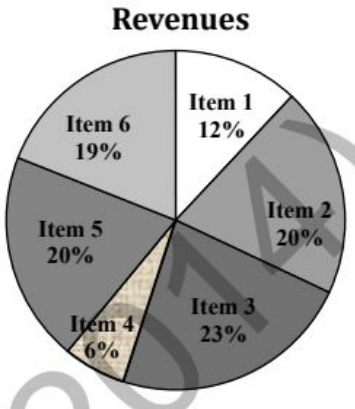
Q.8 The sum of eight consecutive odd numbers is 656. The average of four consecutive even numbers is 87. What is the sum of the smallest odd number and second largest even number?

Answer: 163 to 163

GATE 2014 Set 2 | Q.09

Q.9 The total exports and revenues from the exports of a country are given in the two charts shown below. The pie chart for exports shows the quantity of each item exported as a percentage of the total quantity of exports. The pie chart for the revenues shows the percentage of the total revenue generated through export of each item. The total quantity of exports of all the items is 500 thousand tonnes and the total revenues are 250 crore rupees. Which item among the following has generated the maximum revenue per kg?





(A) Item 2 (B) Item 3 (C) Item 6 (D) Item 5

Answer: D

GATE 2014 Set 2 | Q.10

Q.10 It takes 30 minutes to empty a half-full tank by draining it at a constant rate. It is decided to simultaneously pump water into the half-full tank while draining it. What is the rate at which water has to be pumped in so that it gets fully filled in 10 minutes?

(A) 4 times the draining rate
(B) 3 times the draining rate
(C) 2.5 times the draining rate
(D) 2 times the draining rate

END OF THE QUESTION PAPER

GA04 (GATE

Answer: A

GATE 2014 Set 2 | Q.01

Q.1 The determinant of matrix A is 5 and the determinant of matrix B is 40. The determinant of matrix AB is _____.

Answer: 199 to 201

GATE 2014 Set 2 | Q.02

Q.2 Let X be a random variable which is uniformly chosen from the set of positive odd numbers less than 100. The expectation, $E[X]$, is _____.

Answer: 49.9 to 50.1

GATE 2014 Set 2 | Q.03

Q.3 For $0 \leq t < \infty$, the maximum value of the function $f(t) = e^{-t} - 2\,e^{-2t}$ occurs at

(A) $t = \log_e 4$ (B) $t = \log_e 2$ (C) $t = 0$ (D) $t = \log_e 8$

Answer: A

GATE 2014 Set 2 | Q.04

Q.4 The value of

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{x}$$

is

(A) $ln2$ (B) 1.0 (C) e (D) ∞

Answer: C

GATE 2014 Set 2 | Q.05

Q.5 If the characteristic equation of the differential equation

$$\frac{d^2y}{dx^2}+2\,\alpha\frac{dy}{dx}+y=0$$

has two equal roots, then the values of α are

(A) ±1 (B) 0,0 (C) $\pm j$ (D) ±1/2

Answer: A

GATE 2014 Set 2 | Q.06

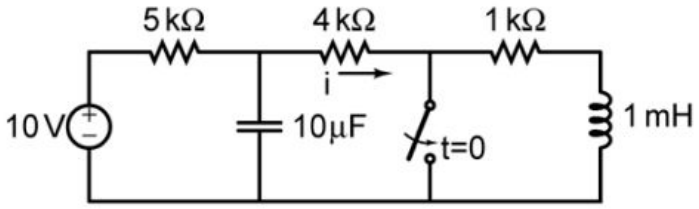
Q.6 Norton's theorem states that a complex network connected to a load can be replaced with an equivalent impedance

(A) in series with a current source
(B) in parallel with a voltage source
(C) in series with a voltage source
(D) in parallel with a current source

Answer: D

GATE 2014 Set 2 | Q.07

Q.7 In the figure shown, the ideal switch has been open for a long time. If it is closed at $t = 0$, then the magnitude of the current (in mA) through the 4 kΩ resistor at $t = 0^+$ is _____.



Answer: 1.2 to 1.3

ECE GATE 2014

GATE 2014 Set 2 | Q.08

Q.8 A silicon bar is doped with donor impurities $N_D = 2.25 \times 10^{15}$ atoms / cm^3. Given the intrinsic carrier concentration of silicon at T = 300 K is $n_i = 1.5 \times 10^{10}$ cm^{-3}. Assuming complete impurity ionization, the equilibrium electron and hole concentrations are

(A) $n_0 = 1.5 \times 10^{16}$ cm^{-3}, $p_0 = 1.5 \times 10^{5}$ cm^{-3}
(B) $n_0 = 1.5 \times 10^{10}$ cm^{-3}, $p_0 = 1.5 \times 10^{15}$ cm^{-3}
(C) $n_0 = 2.25 \times 10^{15}$ cm^{-3}, $p_0 = 1.5 \times 10^{10}$ cm^{-3}
(D) $n_0 = 2.25 \times 10^{15}$ cm^{-3}, $p_0 = 1 \times 10^{5}$ cm^{-3}

Answer: D

GATE 2014 Set 2 | Q.09

Q.9 An increase in the base recombination of a BJT will increase

(A) the common emitter dc current gain β
(B) the breakdown voltage BV_{CEO}
(C) the unity-gain cut-off frequency f_T
(D) the transconductance g_m

Answer: B

GATE 2014 Set 2 | Q.10

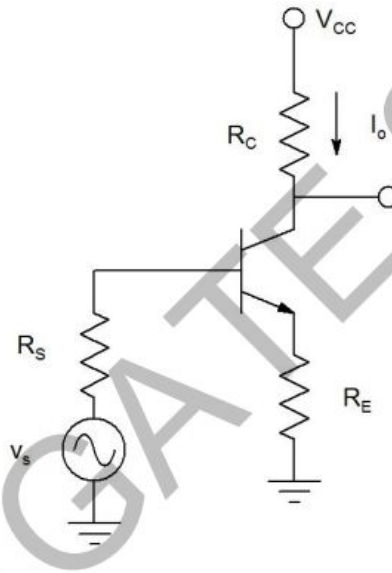
Q.10 In CMOS technology, shallow P-well or N-well regions can be formed using

(A) low pressure chemical vapour deposition
(B) low energy sputtering
(C) low temperature dry oxidation
(D) low energy ion-implantation

Answer: D

GATE 2014 Set 2 | Q.11

Q.11 The feedback topology in the amplifier circuit (the base bias circuit is not shown for simplicity) in the figure is

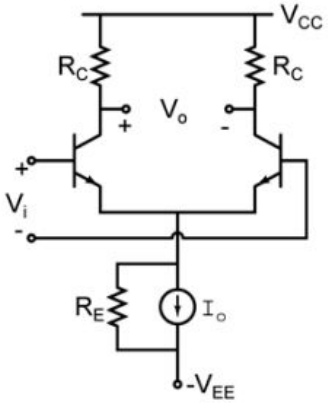


(A) Voltage shunt feedback
(B) Current series feedback
(C) Current shunt feedback
(D) Voltage series feedback

Answer: B

GATE 2014 Set 2 | Q.12

Q.12 In the differential amplifier shown in the figure, the magnitudes of the common-mode and differential-mode gains are A_{cm} and A_d, respectively. If the resistance R_E is increased, then



(A) A_{cm} increases
(B) common-mode rejection ratio increases
(C) A_d increases
(D) common-mode rejection ratio decreases

Answer: B

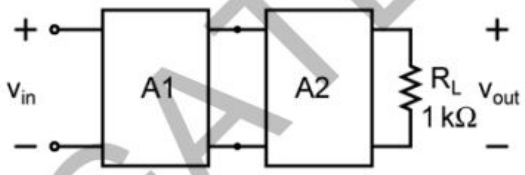
EC02 (GATE 2014)

GATE 2014 Set 2 | Q.13

Q.13 A cascade connection of two voltage amplifiers A1 and A2 is shown in the figure. The open-loop gain A_{v0}, input resistance R_{in}, and output resistance R_o for A1 and A2 are as follows:

A1: $A_{v0} = 10, R_{in} = 10\,k\Omega, R_o = 1\,k\Omega.$
A2: $A_{v0} = 5, R_{in} = 5\,k\Omega, R_o = 200\,\Omega.$

The approximate overall voltage gain v_{out}/v_{in} is _____.



Answer: 34.0 to 35.3

GATE 2014 Set 2 | Q.14

Q.14 For an n-variable Boolean function, the maximum number of prime implicants is

(A) $2(n\text{-}1)$ (B) $n/2$ (C) 2^n (D) $2^{(n-1)}$

Answer: D

GATE 2014 Set 2 | Q.15

Q.15 The number of bytes required to represent the decimal number 1856357 in packed BCD (Binary Coded Decimal) form is _____ .

Answer: 3.9 to 4.1

GATE 2014 Set 2 | Q.16

Q.16 In a half-subtractor circuit with X and Y as inputs, the Borrow (M) and Difference ($N = X - Y$) are given by

(A) $M = X \oplus Y,\ N = XY$
(B) $M = \bar{X}Y,\ N = X \oplus Y$
(C) $M = \bar{X}Y,\ N = X \oplus Y$
(D) $M = X\bar{Y},\ N = \overline{X \oplus Y}$

Answer: C

GATE 2014 Set 2 | Q.17

Q.17 An FIR system is described by the system function

$$H(z) = 1 + \frac{7}{2}z^{-1} + \frac{3}{2}z^{-2}$$

The system is

(A) maximum phase (B) minimum phase (C) mixed phase (D) zero phase

Answer: C

GATE 2014 Set 2 | Q.18

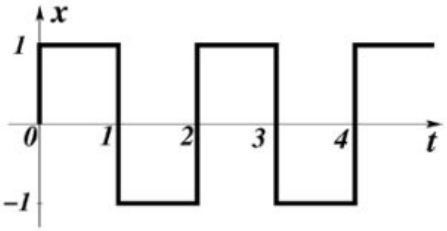
Q.18 Let $x[n] = x[-n]$. Let $X(z)$ be the z-transform of $x[n]$. If $0.5 + j\,0.25$ is a zero of $X(z)$, which one of the following must also be a zero of $X(z)$.

(A) $0.5 - j\,0.25$ (B) $1/(0.5 + j\,0.25)$ (C) $1/(0.5 - j\,0.25)$ (D) $2+j\,4$

Answer: B

GATE 2014 Set 2 | Q.19

Q.19 Consider the periodic square wave in the figure shown.



The ratio of the power in the 7th harmonic to the power in the 5th harmonic for this waveform is closest in value to _____.

Answer: 0.50 to 0.52

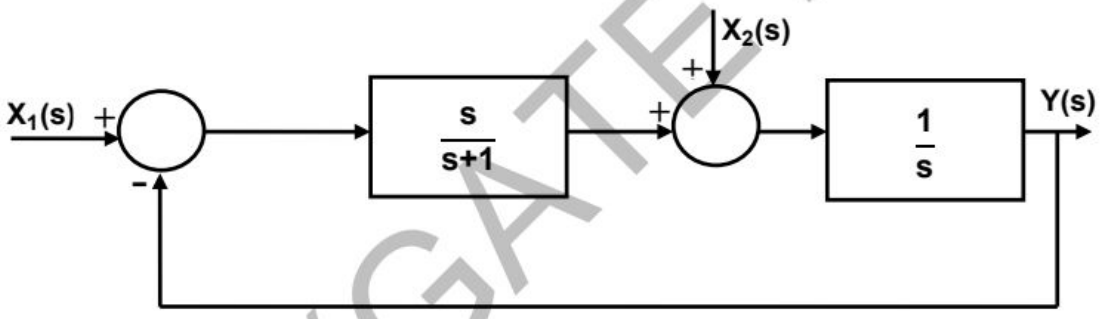
GATE 2014 Set 2 | Q.20

Q.20 The natural frequency of an undamped second-order system is 40 rad/s. If the system is damped with a damping ratio 0.3, the damped natural frequency in rad/s is _____.

Answer: 38.13 to 38.19

GATE 2014 Set 2 | Q.21

Q.21 For the following system,



when $X_1(s) = 0$, the transfer function $\frac{Y(s)}{X_2(s)}$ is

(A) $\frac{s+1}{s^2}$ (B) $\frac{1}{s+1}$ (C) $\frac{s+2}{s(s+1)}$ (D) $\frac{s+1}{s(s+2)}$

Answer: D

GATE 2014 Set 2 | Q.22

Q.22 The capacity of a band-limited additive white Gaussian noise (AWGN) channel is given by $C = W \log_2\left(1 + \frac{P}{\sigma^2 W}\right)$ bits per second (bps), where W is the channel bandwidth, P is the average power received and σ^2 is the one-sided power spectral density of the AWGN.

For a fixed $\frac{P}{\sigma^2} = 1000$, the channel capacity (in kbps) with infinite bandwidth ($W \rightarrow \infty$) is approximately

(A) 1.44 (B) 1.08 (C) 0.72 (D) 0.36

Answer: A

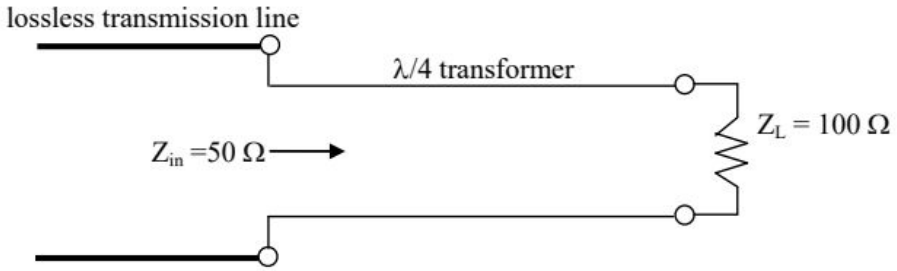
GATE 2014 Set 2 | Q.23

Q.23 Consider sinusoidal modulation in an AM system. Assuming no overmodulation, the modulation index (μ) when the maximum and minimum values of the envelope, respectively, are 3 V and 1 V, is _____.

Answer: 0.45 to 0.55

GATE 2014 Set 2 | Q.24

Q.24 To maximize power transfer, a lossless transmission line is to be matched to a resistive load impedance via a λ/4 transformer as shown.



The characteristic impedance (in Ω) of the λ/4 transformer is _____.

Answer: 70 to 72

GATE 2014 Set 2 | Q.25

Q.25 Which one of the following field patterns represents a TEM wave travelling in the positive x direction?

(A) $E = +8\hat{y},\ H = -4\hat{z}$ (B) $E = -2\hat{y},\ H = -3\hat{z}$
(C) $E = +2\hat{z},\ H = +2\hat{y}$ (D) $E = -3\hat{y},\ H = +4\hat{z}$

Answer: B

GATE 2014 Set 2 | Q.26

Q.26 The system of linear equations

$$\begin{pmatrix} 2 & 1 & 3 \\ 3 & 0 & 1 \\ 1 & 2 & 5 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 5 \\ -4 \\ 14 \end{pmatrix} \text{ has}$$

(A) a unique solution
(B) infinitely many solutions
(C) no solution
(D) exactly two solutions

Answer: B

GATE 2014 Set 2 | Q.27

Q.27 The real part of an analytic function $f(z)$ where $z = x + jy$ is given by $e^{-y}\cos(x)$. The imaginary part of $f(z)$ is

(A) $e^{y}\cos(x)$ (B) $e^{-y}\sin(x)$ (C) $-e^{y}\sin(x)$ (D) $-e^{-y}\sin(x)$

Answer: B

GATE 2014 Set 2 | Q.28

Q.28 The maximum value of the determinant among all 2×2 real symmetric matrices with trace 14 is _____.

Answer: 48.9 to 49.1

GATE 2014 Set 2 | Q.29

Q.29 If $\vec{r} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$ and $|\vec{r}| = r$, then $\text{div}(r^2\nabla(\ln r)) =$ _____.

Answer: 2.9 to 3.1

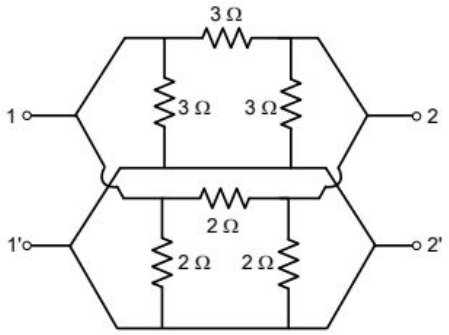
GATE 2014 Set 2 | Q.30

Q.30 A series LCR circuit is operated at a frequency different from its resonant frequency. The operating frequency is such that the current leads the supply voltage. The magnitude of current is half the value at resonance. If the values of L, C and R are 1 H, 1 F and 1 Ω, respectively, the operating angular frequency (in rad/s) is _____.

Answer: 0.45 to 0.47

GATE 2014 Set 2 | Q.31

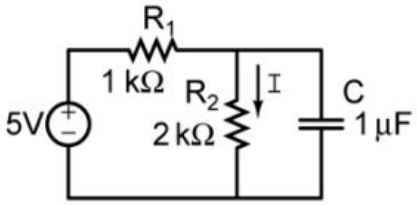
Q.31 In the h-parameter model of the 2-port network given in the figure shown, the value of h_{22} (in S) is _____ .



Answer: 1.24 to 1.26

GATE 2014 Set 2 | Q.32

Q.32 In the figure shown, the capacitor is initially uncharged. Which one of the following expressions describes the current I(t) (in mA) for $t > 0$?



(A) $I(t) = \frac{5}{3}(1 - e^{-t/\tau}),\ \tau = \frac{2}{3}$ msec

(B) $I(t) = \frac{5}{2}(1 - e^{-t/\tau}),\ \tau = \frac{2}{3}$ msec

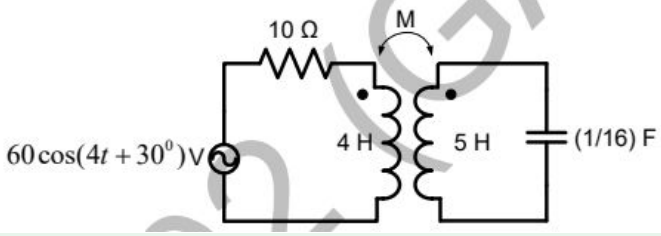
(C) $I(t) = \frac{5}{3}(1 - e^{-t/\tau}),\ \tau = 3$ msec

(D) $I(t) = \frac{5}{2}(1 - e^{-t/\tau}),\ \tau = 3$ msec

Answer: A

GATE 2014 Set 2 | Q.33

Q.33 In the magnetically coupled circuit shown in the figure, 56 % of the total flux emanating from one coil links the other coil. The value of the mutual inductance (in H) is _____ .



Answer: 2.49 to 2.52

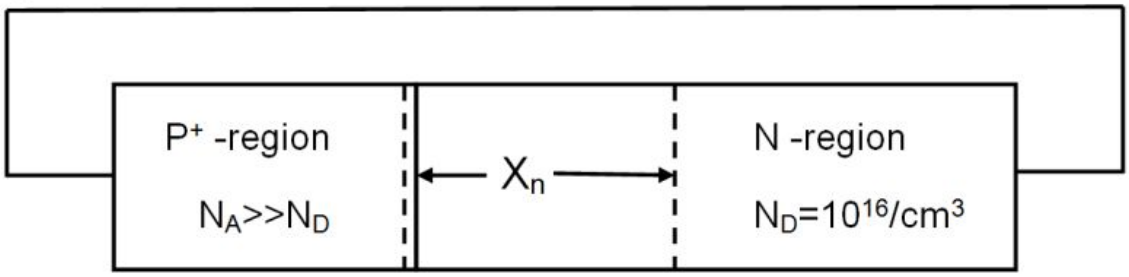
GATE 2014 Set 2 | Q.34

Q.34 Assume electronic charge q = 1.6×10^{-19} C, kT/q = 25 mV and electron mobility μ_n = 1000 cm^2/V-s. If the concentration gradient of electrons injected into a P-type silicon sample is $1\times10^{21}/cm^4$, the magnitude of electron diffusion current density (in A/cm^2) is _____.

Answer: 3990 to 4010

GATE 2014 Set 2 | Q.35

Q.35 Consider an abrupt PN junction (at T = 300 K) shown in the figure. The depletion region width X_n on the N-side of the junction is 0.2 μm and the permittivity of silicon (ε_{si}) is 1.044×10^{-12} F/cm. At the junction, the approximate value of the peak electric field (in kV/cm) is _____.



Answer: 30 to 32

GATE 2014 Set 2 | Q.36

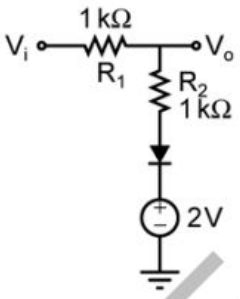
Q.36 When a silicon diode having a doping concentration of $N_A = 9 \times 10^{16}$ cm^{-3} on p-side and $N_D = 1 \times 10^{16}$ cm^{-3} on n-side is reverse biased, the total depletion width is found to be 3 μm. Given that the permittivity of silicon is 1.04×10^{-12} F/cm, the depletion width on the p-side and the maximum electric field in the depletion region, respectively, are

(A) 2.7 μm and 2.3×10^5 V/cm
(B) 0.3 μm and 4.15×10^5 V/cm
(C) 0.3 μm and 0.42×10^5 V/cm
(D) 2.1 μm and 0.42×10^5 V/cm

Answer: B

GATE 2014 Set 2 | Q.37

Q.37 The diode in the circuit shown has $V_{on} = 0.7$ Volts but is ideal otherwise. If $V_i = 5\sin(\omega t)$ Volts, the minimum and maximum values of V_o (in Volts) are, respectively,

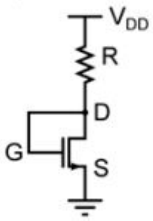


(A) −5 and 2.7
(B) 2.7 and 5
(C) −5 and 3.85
(D) 1.3 and 5

Answer: C

GATE 2014 Set 2 | Q.38

Q.38 For the n-channel MOS transistor shown in the figure, the threshold voltage V_{Th} is 0.8 V. Neglect channel length modulation effects. When the drain voltage $V_D = 1.6$ V, the drain current I_D was found to be 0.5 mA. If V_D is adjusted to be 2 V by changing the values of R and V_{DD}, the new value of I_D (in mA) is

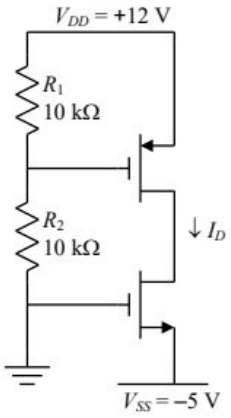


(A) 0.625 (B) 0.75 (C) 1.125 (D) 1.5

Answer: C

GATE 2014 Set 2 | Q.39

Q.39 For the MOSFETs shown in the figure, the threshold voltage $|V_t| = 2$ V and $K = \frac{1}{2}\mu C_{ox}\left(\frac{W}{L}\right) = 0.1$ mA/V^2. The value of I_D (in mA) is _____ .

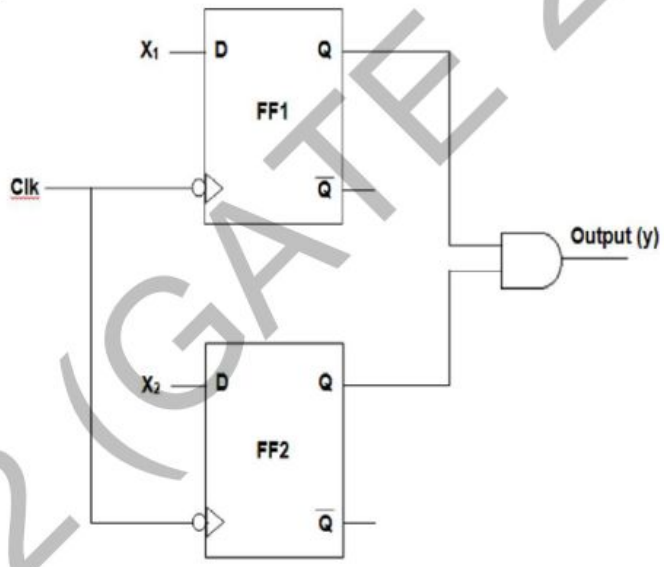


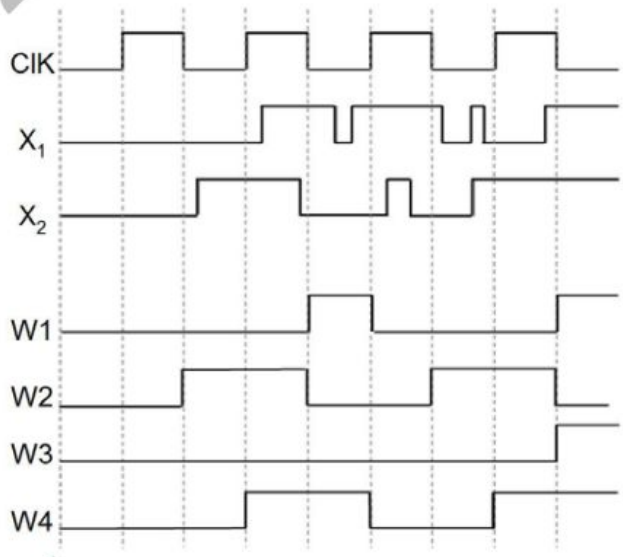
Answer: 0.88 to 0.92

EC02 (GATE 2014)

GATE 2014 Set 2 | Q.40

Q.40 In the circuit shown, choose the correct timing diagram of the output (y) from the given waveforms W1, W2, W3 and W4.





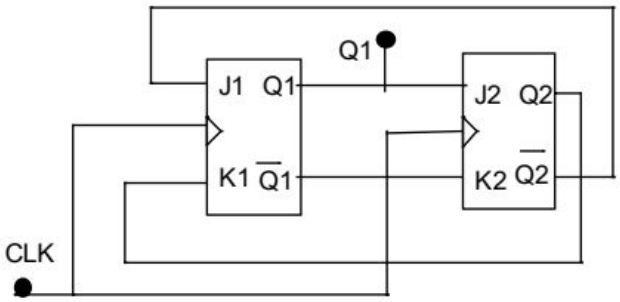
(A) W1 (B) W2 (C) W3 (D) W4

EC02 (GATE 2

Answer: C

GATE 2014 Set 2 | Q.41

Q.41 The outputs of the two flip-flops Q1, Q2 in the figure shown are initialized to 0, 0. The sequence generated at Q1 upon application of clock signal is

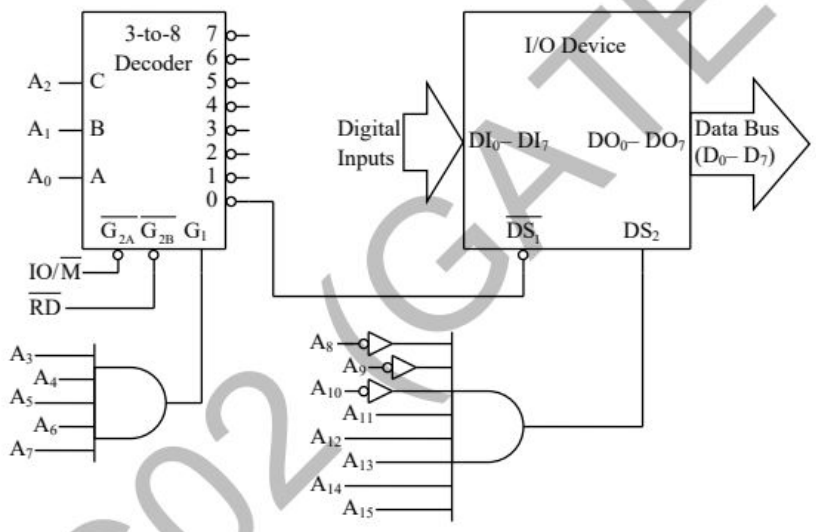


(A) 01110…
(B) 01010…
(C) 00110…
(D) 01100…

Answer: D

GATE 2014 Set 2 | Q.42

Q.42 For the 8085 microprocessor, the interfacing circuit to input 8-bit digital data ($DI_0 - DI_7$) from an external device is shown in the figure. The instruction for correct data transfer is



(A) MVI A, F8H (B) IN F8H (C) OUT F8H (D) LDA F8F8H

Answer: D

GATE 2014 Set 2 | Q.43

Q.43 Consider a discrete-time signal

$$x[n] = \begin{cases} n & \text{for } 0 \le n \le 10 \\ 0 & \text{otherwise} \end{cases}.$$

If $y[n]$ is the convolution of $x[n]$ with itself, the value of $y[4]$ is _____.

Answer: 9.9 to 10.1

EC 02 (GATE 2014)

GATE 2014 Set 2 | Q.44

Q.44 The input-output relationship of a causal stable LTI system is given as

$$y[n] = \alpha\, y[n-1] + \beta\, x[n]$$

If the impulse response $h[n]$ of this system satisfies the condition $\sum_{n=0}^{\infty} h[n] = 2$, the relationship between α and β is

(A) $\alpha = 1 - \beta/2$ (B) $\alpha = 1 + \beta/2$ (C) $\alpha = 2\beta$ (D) $\alpha = -2\beta$

Answer: A

GATE 2014 Set 2 | Q.45

Q.45 The value of the integral $\int_{-\infty}^{\infty} \text{sinc}^2(5t)\, dt$ is _____.

Answer: 0.19 to 0.21

GATE 2014 Set 2 | Q.46

Q.46 An unforced linear time invariant (LTI) system is represented by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

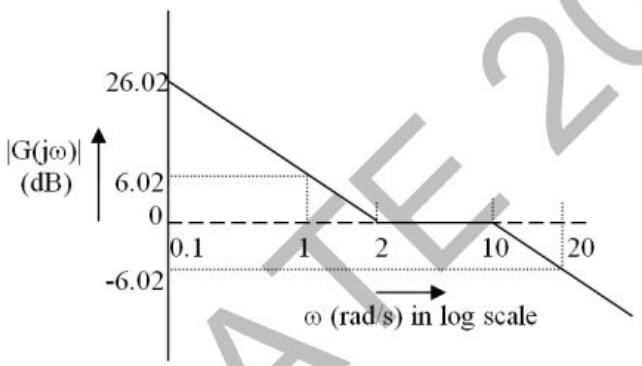
If the initial conditions are $x_1(0) = 1$ and $x_2(0) = -1$, the solution of the state equation is

(A) $x_1(t) = -1,\ x_2(t) = 2$
(B) $x_1(t) = -e^{-t},\ x_2(t) = 2e^{-t}$
(C) $x_1(t) = e^{-t},\ x_2(t) = -e^{-2t}$
(D) $x_1(t) = -e^{-t},\ x_2(t) = -2e^{-t}$

Answer: C

GATE 2014 Set 2 | Q.47

Q.47 The Bode asymptotic magnitude plot of a minimum phase system is shown in the figure.



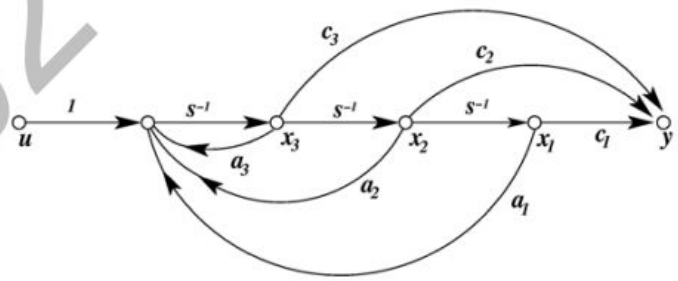
If the system is connected in a unity negative feedback configuration, the steady state error of the closed loop system, to a unit ramp input, is _____.

Answer: 0.49 to 0.51

GATE 2014

GATE 2014 Set 2 | Q.48

Q.48 Consider the state space system expressed by the signal flow diagram shown in the figure.



The corresponding system is

(A) always controllable
(B) always observable
(C) always stable
(D) always unstable

Answer: A

GATE 2014 Set 2 | Q.49

Q.49 The input to a 1-bit quantizer is a random variable X with pdf $f_X(x) = 2e^{-2x}$ for $x \geq 0$ and $f_X(x) = 0$ for $x < 0$. For outputs to be of equal probability, the quantizer threshold should be _____.

Answer: 0.34 to 0.36

GATE 2014 Set 2 | Q.50

Q.50 Coherent orthogonal binary FSK modulation is used to transmit two equiprobable symbol waveforms $s_1(t) = \alpha \cos 2\pi f_1 t$ and $s_2(t) = \alpha \cos 2\pi f_2 t$, where $\alpha = 4$ mV. Assume an AWGN channel with two-sided noise power spectral density $\frac{N_0}{2} = 0.5 \times 10^{-12}$ W/Hz. Using an optimal receiver and the relation $Q(v) = \frac{1}{\sqrt{2\pi}} \int_v^{\infty} e^{-u^2/2}\, du$, the bit error probability for a data rate of 500 kbps is

(A) $Q(2)$ (B) $Q(2\sqrt{2})$ (C) $Q(4)$ (D) $Q(4\sqrt{2})$

Answer: C

GATE 2014 Set 2 | Q.51

Q.51 The power spectral density of a real stationary random process $X(t)$ is given by

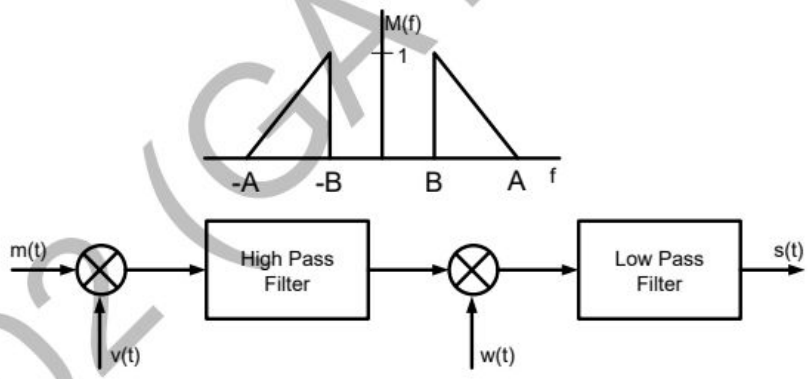
$$S_X(f) = \begin{cases} \frac{1}{W}, & |f| \leq W \\ 0, & |f| > W \end{cases}.$$

The value of the expectation $E\left[\pi X(t)\, X\left(t - \frac{1}{4W}\right)\right]$ is _____.

Answer: 3.9 to 4.1

GATE 2014 Set 2 | Q.52

Q.52 In the figure, $M(f)$ is the Fourier transform of the message signal $m(t)$ where A = 100 Hz and B = 40 Hz. Given $v(t) = \cos(2\pi f_c t)$ and $w(t) = \cos(2\pi(f_c + A)t)$, where $f_c > A$. The cutoff frequencies of both the filters are f_c.



The bandwidth of the signal at the output of the modulator (in Hz) is _____.

Answer: 59.9 to 60.1

GATE 2014 Set 2 | Q.53

Q.53 If the electric field of a plane wave is

$$\vec{E}(z,t) = \hat{x}3\cos(\omega t - kz + 30°) - \hat{y}4\sin(\omega t - kz + 45°)\ (\text{mV/m}),$$

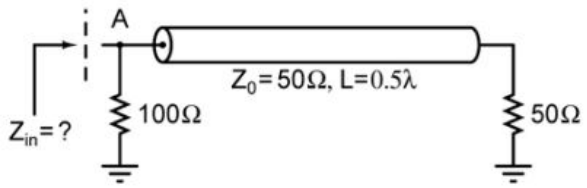
the polarization state of the plane wave is

(A) left elliptical
(B) left circular
(C) right elliptical
(D) right circular

Answer: A

GATE 2014 Set 2 | Q.54

Q.54 In the transmission line shown, the impedance Z_{in} (in ohms) between node A and the ground is _____.



Answer: 32.99 to 34.01

GATE 2014 Set 2 | Q.55

Q.55 For a rectangular waveguide of internal dimensions $a \times b$ $(a > b)$, the cut-off frequency for the TE_{11} mode is the arithmetic mean of the cut-off frequencies for TE_{10} mode and TE_{20} mode. If $a = \sqrt{5}$ cm, the value of b (in cm) is _____.

END OF THE QUESTION PAPER

EC02 (GATE 2014)

Answer: 1.9 to 2.1